
CS@SIT Hackathon 2026 Project Details and Description

CSC105 Web Application Development (2/2025)

Project Title: SolarSafe: Sun Exposure Management and Survival Planning Web Application

Product Title: SolarSafe

Overview:

The world has already entered an apocalyptic era. Due to extreme global warming and the collapse of the Earth's atmospheric protection, solar radiation reaching the surface has become deadly. Even a few minutes of direct exposure to sunlight can cause severe skin burns, vision damage, and other life-threatening conditions. Most people now live indoors and only travel at night, while those who must go outside during the day face significant risks.

The main problem is that people still need to leave shelter for essential activities such as medical emergencies, critical work, and resource collection, but they have no reliable way to determine how long they can safely remain outdoors. Existing planner applications do not account for dangerous environmental conditions or UV exposure limits.

SolarSafe was developed to help people survive in this harsh environment. The application calculates safe outdoor exposure times, helps users plan and schedule activities, and recommends appropriate protective equipment based on current sunlight conditions. By combining a planner, exposure calculator, and personalized safety recommendations, SolarSafe enables users to make informed decisions and reduce the risk of deadly sun exposure.

Assessable Features:

F1 [Distinct]: Planner

Responsible Member(s): [858] [Chanakarn]

- Users can create, edit, and delete outdoor activities.
- Users can view their daily schedule through a centralized planner dashboard.
- The system displays color-coded UV risk levels for different time periods.
- Users can organize activities around safer outdoor time windows.
- Users can view all scheduled activities and manage their daily survival plan.

F2 [Support]: Sun Safety Calculator

Responsible Member(s): [868] [Saranpat]

- Users can calculate safe outdoor exposure time based on skin type.
- Users can select protective equipment to increase safe exposure duration.
- The system automatically updates safety calculations when inputs change.
- Users can start and monitor a countdown timer during outdoor activities.
- Users can reset calculations and timers when needed.

F3 [Support]: Protective Equipment Recommendation System

Responsible Member(s): [855] [Jomnaree]

- Users can browse available protective equipment by category.
- Users can filter equipment by price range and protection level.
- Users can view equipment details including protection score and cost.
- Users can add recommended equipment to their protection loadout.
- Selected equipment is integrated with the Sun Safety Calculator to improve safety recommendations.